

SAP Leave Management System

Backend Project Documentation | SAP ABAP Cloud

1. Project Overview

A backend-focused SAP Leave Management System developed using SAP ABAP Cloud. It manages employees, leave types, leave balances and leave requests, with validation, manager approval/rejection, balance updates and reporting.

2. Architecture

SAP ABAP Cloud → Custom Tables → ABAP Business Logic → Reports / OData V4

The application is implemented in Eclipse using ABAP Development Tools. No React/Fiori UI or external authentication integration is required for the current project scope.

3. Database Tables

Table	Purpose
ZLM_EMPLOYEE	Employee master: ID, name, department, designation, email and join date.
ZLM_LEAVE_TYPE	Leave master: CL, SL and EL with allowed days.
ZLM_LEAVE_BAL	Employee leave balances for CL, SL and EL.
ZLM_LEAVE_REQ	Leave requests, dates, reason, status, manager and balance-update flag.

4. ABAP Classes

Class	Responsibility
ZCL_LEAVE_SAMPLE_DATA	Inserts sample employees, leave types and initial balances.
ZCL_LEAVE_VALIDATION	Validates employee, dates and balance; creates PENDING requests.
ZCL_LEAVE_APPROVAL	Approves only PENDING requests.
ZCL_LEAVE_REJECTION	Rejects only PENDING requests without reducing balance.
ZCL_LEAVE_BALANCE_UPDATE	Deducts approved leave and prevents duplicate deductions.
ZCL_LEAVE_REPORT	Displays leave history and request status.

5. Business Rules

- Employee must exist before applying leave.
- From Date must not be after To Date.
- Requested days must not exceed the available balance.
- Valid requests are stored as PENDING.
- Only PENDING requests can be approved or rejected.
- Rejected requests do not reduce leave balance.
- Only APPROVED requests can update leave balance.
- BALANCE_UPDATED = 'X' prevents duplicate balance deduction.

6. Request Lifecycle

Apply Leave → Validate → PENDING → Manager Decision → APPROVED/REJECTED → Balance Update if Approved → Report

7. Leave Types

Code	Leave Type	Example Initial Balance
CL	Casual Leave	12 days
SL	Sick Leave	10 days
EL	Earned Leave	15 days

8. OData Service

The backend exposes leave-request data through the OData V4 service **ZUI_LM_LEAVE**. The service exposes the **LeaveRequests** entity set based on the CDS view **ZC_LM_LEAVE_REQ**.

9. Testing Completed

- Valid CL, SL and EL requests created successfully.
- Non-existent employee rejected.
- Invalid date range rejected.
- Insufficient balance rejected.
- Pending request approved successfully.
- Pending request rejected successfully.
- Repeated approval/rejection blocked.
- Approved leave balance deducted correctly.
- Repeated balance deduction blocked.
- Leave history report executed successfully.

10. Interview Explanation

“My project is an SAP Leave Management System developed using ABAP Cloud. I created custom tables for employees, leave types, leave balances and leave requests. I implemented ABAP validation for employee existence, leave dates and available balance. Valid requests enter **PENDING** status and can then be approved or rejected by a manager. Approved requests reduce the corresponding leave balance, while rejected requests do not. I also implemented duplicate balance-update protection, leave history reporting and an OData V4 service.”

11. Current Scope

The project is intentionally backend-only. The completed scope consists of the SAP database model, ABAP business logic, approval/rejection processing, balance management, reporting and OData exposure. A UI and external communication integration are outside the current scope.

Project Status: Backend MVP Complete